

Graphs and Complexity for IoT

Master IoT

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Lecture 3

- ▶ Computational complexity classes P , NP , NP -Completeness, NP -Hardness
- ▶ $P = NP$? Conjecture
- ▶ Famous hard problems
- ▶ Approximation algorithms
- ▶ Graph coloring, vertex and edge covering, domination in graphs, maximum clique, set covering, TSP problem

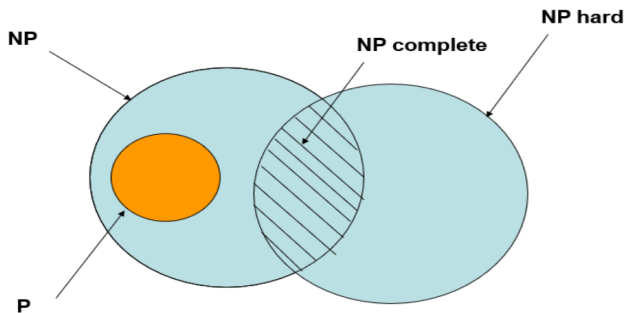
Reminder of the complexity class values:

- ▶ $\mathcal{O}(\log 1)$: constant
- ▶ $\mathcal{O}(\log n)$: logarithmic
- ▶ $\mathcal{O}(n)$: linear
- ▶ $\mathcal{O}(n \log n)$: sub-quadratic
- ▶ $\mathcal{O}(n^k)$: polynomial
- ▶ $\mathcal{O}(a^n)$: exponential
- ▶ $\mathcal{O}(n!)$: factorial

Complexity classes:

- ▶ **P**: Polynomial Time class
- ▶ **NP**: Non Deterministic Polynomial Time class
- ▶ **NP-Complete & NP-Hard**
- ▶ **$P=NP$?**
One of the Millennium Prize Problems - Clay Mathematics Institute

Complexity classes



Some hard problems:

- ▶ **Subset Sum problem:** whether or not a subset from a list of integers can sum to a target value.
- ▶ **TSP:** find the shortest possible route that visits a given set of cities and returns to the starting city
- ▶ **Clique problem:** does a graph G have a clique of size at least K ?
- ▶ **Task scheduling problem:** a set of tasks, a set of machines, minimise the time makespan.
- ▶ **Set covering problem:** given a pair $(\mathcal{U}, \mathcal{S})$, and the objective is to find a set cover that uses the fewest sets.

Approximation algorithms:

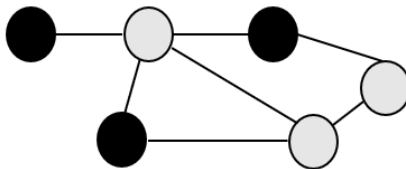
- ▶ When finding optimal solutions is intractable
- ▶ An approximation algorithm gives a solution to a combinatorial optimization problem which is provably close to optimal
- ▶ An algorithm is factor α approximation (α -approximation algorithm) for a problem iff for every instance of the problem it can find a solution within a factor α of the optimum solution.
- ▶ minimization ($\alpha > 1$): this definition implies that the solution found by the algorithm is at most α times the optimum solution.
- ▶ maximization ($\alpha < 1$): this definition guarantees that the approximate solution is at least α times the optimum.

Approximation algorithms - Examples:

- ▶ TSP problem: $3/2$ - approximation (Christofides)
- ▶ Task scheduling:
 - ▶ SPT algorithm: $2 - 1/m$ - approximation
 - ▶ LPT algorithm: $4/3 - 1/3m$ - approximation
- ▶ Set covering problem: $\log n$ - approximation (greedy algorithm)

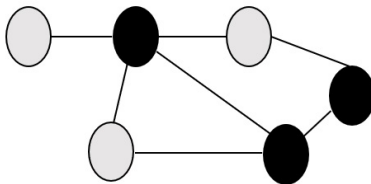
Some nice problems from graph theory:

1- **Independent set:** A set of vertices where no two vertices are adjacent



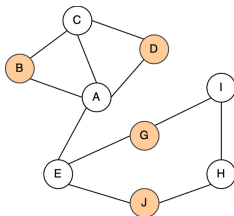
Some nice problems from graph theory:

2- **Vertex Cover**: it is a set of nodes that cover all the edges in the graph

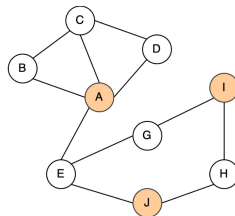


Some nice problems from graph theory:

3- Dominating set : a dominating set for $G = (V, E)$ is a subset D of V st. every vertex not in D is adjacent to at least one in D .



(a) A minimal dominating set D_1 .



(b) A minimum dominating set D_2 .

Some nice problems from graph theory:

Graph colouring : the problem of coloring nodes of a graph G in such a way that no two adjacent vertices have the same color.

Chromatic number $\chi(G)$: the minimum number of colors.

